

Project

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1 Introduction

1.1 Project Goal

This paper presents the development and evaluation of an educational conversational agent specified to women’s mental health, utilizing a Retrieval-Augmented Generation (RAG) framework. The primary objective is to conduct a comparative analysis of various RAG architectures, specifically single-agent RAG, multi-agent RAG, and question-to-question RAG, as well the integration of a symptom graph, to assess their respective impacts on system efficacy and response generation. To contextualize this work, the remainder of the introduction establishes the critical need for specialized mental health support tools for women and highlights the inherent advantages of applying RAG frameworks to this domain. Following this, the related work section provides a comprehensive review of existing literature and prior research. The methodology then details the theoretical foundations, key terminology, and technical design choices driving the proposed architectures. Subsequently, the Results and Evaluation section presents a detailed comparative analysis of the model outputs, discussing current system limitations and potential avenues for future improvement. Finally, the conclusion synthesizes the core contributions and primary findings of this research.

1.2 Importance of Women’s Mental Health

1.2.1 Social dynamic crisis

The landscape of global mental health has shifted drastically over the past years. According to the recent statistics from World Health Organization, there are around 1.2 billion people that are living with recognised mental health condition. This makes mental illness one of the leading drivers of global disability. The increase of people with these issues is strongly correlated with rising problems in social dynamics, most notably the psychological effects of COVID-19 pandemics.

The post pandemic era had major influence on traditional communication patterns. Extended periods of isolation have created strong reliance on digital first interactions, which caused widespread social anxiety. For many people navigating face to face interactions became stressful task, driving away people from physical support and pushing them to unhealthy coping strategies.

1.2.2 Economic crisis

However, human crisis is not the only problem when dealing with global mental health problem. Globally, there is a huge mismatch between the demand for the psychological support and economic

resources that are allocated for it.

Massive macroeconomic crisis happens with low productivity, absenteeism, premature economic exit, which are often consequences of mental health struggles. In addition, despite the fact that globally around 8% of medical problems are related to psychological issues, only 2% of the overall health budget is allocated to mental health care, which is almost unchanged for almost a decade.

The systematic underfunding leaves majority of people without accessible help. In high income countries waiting list for clinical psychological evaluations routinely goes from six months to over a year. In low-to-middle income nations, the ratio of professionals to citizens drops significantly to as few as one psychiatrist per million people.

1.2.3 The Gender Disparity in Medical Research

In the current system, there is evident neglect in women’s mental health. Women are diagnosed with generalized anxiety disorder and major depressive disorders at twice the rate of men. Furthermore, younger women aged 16–24 are nearly three times as likely to experience an acute mental health issue as males of the same age. Despite this clear epidemiological reality, medical and psychological research has historically suffered from a profound male bias, creating a structural data deficit that leaves women at a distinct disadvantage when seeking help. This data gap is driven by three main historical and structural factors:

- Systematic trial exclusion that was present until mid 1990s, that happened because of the assumption that hormones would create noisy data.
- Data exploration bias: Because primary datasets were constructed using male subjects, the clinical baselines for diagnosis and pharmaceutical treatments used today are largely extrapolated from male physiology and psychology.
- Critical endocrine milestones unique to female biology, such as the postpartum phase, the premenstrual matrix, and perimenopause are drastically under researched, frequently leaving patients with answers that overlook the intersection of hormonal changes and neurological wellness.

1.3 Combining RAG with mental health domain

1.3.1 Why RAG over standard LLM?

As a consequence of clinical bottlenecks and research gaps, individuals are increasingly using generative AI architectures for support. However, standard Large Language Models (LLMs) present severe risks in sensitive domains, frequently generating hallucinations, or delivering unverified medical advice. To tackle this major issue, Retrieval-Augmented Generation (RAG) introduces a strict architectural constraint. Instead of allowing an LLM to generate text solely from its internal weights, a RAG system forces the model to act as educated dashboard of a verified, external reference library. When a user submits a query, the system first passes it to a semantic vector database containing curated, open-source wellness data. The system retrieves the most relevant text chunks and uses them as a contextual ground for the Large Language Model. The LLM then generates a response that is strictly bound by these documents, completely avoiding medical prescriptions or the simulation of personal relationships.

General benefits of using RAG architecture are reducing the risk of hallucinations that is very common in LLMs, traceable source citations and cost-efficiency compared to fine-tuning, that can cost thousands of dollars.

When applied to women’s mental health, a RAG framework allows a model to bypass the historical

biases of mainstream medicine by building a curated, high-integrity knowledge base drawn directly from open-source lifestyle, nutritional, and psychoeducational literature. Crucially, because this system operates at the intersection of lifestyle adjustments, such as soundscapes, meditation preferences, somatic exercises, and endocrine-conscious nutritional adjustments, it sidesteps the clinical dangers of automated medical prescriptions. The architecture ensures complete user anonymity and provides free access to wellbeing protocols.

1.3.2 Establishing educational non-relational boundary

Important part of this project is the rejection of the artificial or therapeutic personal relationships. Unlike conversational agents that attempt to simulate human empathy or act as artificial friends, this project is designed strictly for educational and knowledge-delivery purposes. The system treats the user not as a patient in a clinical relationship, but as a self-directed agent seeking high-integrity, evidence-based wellness data. By positioning the interface as an interactive reference dashboard rather than an intimate chat companion, the project eliminates the ethical hazard of misleading vulnerable users into believing they are conversing with an emotional entity. It focuses strictly on providing an objective opportunity for knowledge sharing and self-guided improvement.

1.3.3 Existing wellbeing chatbots

Current leading paradigms in this digital mental health space are Woebot and Wysa. While both models gave successful access to psychological tools, there are gaps in operational and philosophical areas that RAG architecture aims to improve.

Historically, Woebot operates on a deterministic, rules based architecture rather than a fully generative one. While it utilizes Natural Language Processing (NLP) at the entry point, it uses this technology purely for input classification, mapping the user's words to specific intent categories. Once the intent is classified, the system routes the user down a pre scripted, human-authored conversational branching path rooted in Cognitive Behavioral Therapy (CBT). The primary limitation of these fixed scripts is that because every response originates from a static folder of pre-approved statements, the interaction can quickly feel repetitive. If a user's queries falls outside the pre-programmed tree, the illusion of understanding breaks down, leading to a mechanical and frustrating user experience that cannot adapt to highly individualized lifestyle queries.

Wysa was traditionally built using a similar hybrid model, combining structured AI guided paths with human-in-the-loop coaching. In recent iterations, the platform has begun integrating clinical Large Language Models to handle complex user inputs and draft weekly reflection summaries. However, because it operates as a commercial, clinical-first application, its main design goal remains relational. It explicitly seeks to establish a persistent digital therapeutic alliance and companion-oriented ongoing check-ins to build emotional compliance.

Architecture of RAG represents a significant evolutionary step forward from both rigid rules-based chatbots and unconstrained commercial conversational agents. It successfully combines the strengths of natural language expression with absolute factual safety across three distinct technical are: First, regarding conversational flexibility, rules-based systems like traditional Woebot are limited to static branching paths, and unconstrained commercial systems present massive safety risks. The RAG architecture offers a dynamic yet bounded approach, where natural language generation remains fluid but completely pinned to external texts. Second, regarding the user relationship goal,

while existing market seeks to build continuous companion-oriented bonds or digital relationships, this project implements a strictly educational model. It functions purely as an objective, high-integrity knowledge-sharing portal. Third, regarding risk management, RAG mitigates the high user friction and robotic repetition found in rules-based systems, while simultaneously reducing the data hallucinations, prompt injections, or accidental medical diagnoses common to standard generative configurations. By transitioning away from rigid, pre-scripted buttons and moving toward a RAG architecture, this project provides users with a fluid, natural language interface that can answer highly specific, granular wellness questions. Simultaneously, by sourcing its knowledge directly from an explicit database of open-source literature on nutrition, somatic exercise, and mindfulness, it guarantees that the output remains entirely factual, traceably cited, and strictly non-medical.

2 Background

- Work that is connected to my thesis.
- Explanation of read research papers

2.1 Technical Foundations and Terminology

2.1.1 Large Language Model (LLM)

A large language model (LLM) is a neural network language model trained to predict and generate sequences of tokens. Most contemporary LLMs use the Transformer architecture, in which attention mechanisms explores parallel relationships between every token pair, including first and last token. This mechanism is allowing the model to capture highly complex relationships. A model’s parameters contain statistical regularities learned during training, but they do not constitute a verified or continuously updated medical database. Consequently, a fluent answer may be outdated, unsupported, or false. In this thesis, the LLM is treated as a controlled language generation component rather than central figure.

2.1.2 RAG

Retrieval-augmented generation (RAG) combines parametric memory in a language model with non-parametric memory stored in an external corpus. [1] introduced the influential RAG formulation for knowledge-intensive Natural Language Processing (NLP) tasks. In a practical application, documents are divided into chunks, encoded as vectors, stored with metadata, and retrieved in response to a user query. The retrieved passages are inserted into the generation prompt so that the model can produce an answer from explicit evidence. RAG supports source updating and provenance without retraining the generator; however, it does not guarantee factuality [2]. Retrieval can miss the best passage, irrelevant evidence can be selected, and a generator can still misuse correct evidence [3]. Health-care reviews therefore describe RAG as a reliability-enhancing method that still requires source validation, evaluation, and human governance [4].

2.1.3 Multi-Agent RAG

Multi-agent RAG separates the workflow into role-bounded components. An agent is not a separate model. In this implementation, an agent is a unit defined by the system prompt, an input and output schema, permitted tools, a domain-specific retrieval filter, a confidence rule, and a stop or escalation condition. Several agents may share one locally hosted instruction-tuned LLM while behaving differently because they receive different evidence and constraints. This design reduces hardware requirements and creates auditable intermediate outputs. Surveys of LLM-based multi-agent systems identify role specialization, communication, and coordination as principal design dimensions, but also note cost, error propagation, and evaluation challenges (?).

2.1.4 Question-To-Question Semantic Retrieval

Standard RAG compares the user query with document or transcript passages. In question-to-question retrieval, the user query is compared with previously prepared canonical questions. Each stored question is connected to a verified answer and one or more supporting transcript spans. The approach is supported by work on question-answer database retrieval, in which a new question is matched against a collection of existing question-answer records (Campese et al., 2023). Verified podcast transcripts, hosted by PhD researchers, will first be divided by topic, speaker turn and timestamp. Candidate Q&A records will be extracted from explicit interview questions and their responses or generated from self-contained evidence spans. Each record will contain a canonical question, optional paraphrases, a concise answer, the exact supporting transcript span, episode identifier, speaker, timestamp, domain labels, population, evidence tier and verification status. The question will be embedded for semantic retrieval, while the answer and evidence span will be stored as linked payload. Podcast studies show that automatic transcripts can support search and summarization but also contain recognition errors, disfluencies and highly variable spoken structures (Clifton et al., 2020). However, the transcripts from Huberman Lab podcast, whose episodes will mainly be used, are human reviewed and verified. In addition, the podcast is led by Andrew Huberman, Ph.D., a neuroscientist and tenured professor in the department of neurobiology, psychiatry and behavioral sciences at Stanford School of Medicine. All of his episodes that will be used in this project are discussion between him and other people with Ph.D. This make the use of podcast transcripts highly reliable.

When a user submits a query, the system will create a query embedding and retrieve the nearest stored questions using cosine similarity. The top candidates will then be reranked using both query-to-question similarity and query-to-answer relevance. A minimum threshold will be required before an answer is released. If no sufficiently similar and answer-relevant record is found, the system will abstain or fall back to passage retrieval. The final response must cite the transcript timestamp and, for medical claims, supporting peer-reviewed or official evidence. Grounding generated content in specific transcript regions is important because transcription errors and unsupported abstraction can introduce factual inconsistencies (Song et al., 2022).

2.1.5 Retrieval Comparison

The question-to-question method is not a replacement for multi-agent orchestration because the two operate at different architectural levels. Multi-agent RAG determines which specialists process

the query and how their outputs are combined. Question-to-question retrieval determines how evidence is found. It can therefore be implemented as an alternative retrieval tool used by the same agents. This project compares: single-agent passage RAG; multi-agent passage RAG; multi-agent question-to-question retrieval; and a graph expansion model.

The language model, agent definitions, prompts, metadata filters and answer budget should remain constant between experimental conditions. Retrieval performance will be measured using Recall@k, mean reciprocal rank and normalized discounted cumulative gain. Answer evaluation should measure factual correctness, citation support, completeness, safety, abstention accuracy, latency and token consumption. A manually reviewed test set should contain direct matches, paraphrases, multi-domain questions, insufficient-evidence questions and safety-sensitive scenarios.

2.1.6 The knowledge graph

The knowledge graph is a typed, attributed multigraph. Nodes represent symptoms, symptom clusters, conditions or educational topics, reproductive life stages, risk and protective factors, interventions, contraindications, evidence sources, and contextual variables. Edges represent qualified relations such as co-occurs-with, may-worsen, may-improve-with, is-associated-with, has-red-flag, contraindicated-for, and supported-by-source. The graph is not a diagnostic causal model. It is an evidence-navigation and ranking structure whose edges retain source identifiers and confidence values. GraphRAG research illustrates how graphs can support multi-hop and global evidence aggregation beyond isolated nearest-neighbor chunks (?).

Table 1: Terminology used in the proposed machine-learning framework.

Term	Technical meaning
Token	A word fragment or symbol processed by the model. Token count determines context and generation length.
Context window	The maximum number of tokens the model can process in one request, including instructions, evidence, history, and output.
Embedding	A dense numeric vector representing semantic content so that related queries and passages can be compared.
Chunk	A bounded passage extracted from a source while retaining page, section, source, and license metadata.
Vector store	An index that stores embeddings and returns passages whose vectors are similar to a query vector.
Sparse retrieval	Term-based retrieval such as BM25, useful for exact symptoms, abbreviations, drug names, and uncommon terminology.
Dense retrieval	Embedding-based retrieval that captures semantic similarity even when query and source use different words.
Hybrid retrieval	Combination of sparse and dense rankings to improve coverage across exact and semantic matches.
Reranker	A slower model that scores query-passage pairs after initial retrieval and reorders the candidate evidence.

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Table 1 continued

Term	Technical meaning
Grounding	Constraining an answer to claims supported by retrieved evidence and traceable citations.
Hallucination	Generated content that is unsupported, incorrect, or produced despite sounding plausible.
Prompt	Instructions and context supplied to the model, including role, evidence, output format, and safety rules.
Structured output	A validated object, normally JSON/Pydantic, with fixed fields rather than unrestricted prose.
Agent state	The shared record containing the query, route, evidence, graph paths, outputs, uncertainty, and safety flags.
Provenance	The source, page or timestamp, passage identifier, extraction method, and model version behind a claim or edge.
Confidence/calibration	A score and evaluation process indicating how often predictions at a stated confidence level are correct.
Precision and recall	Precision measures how many selected items are relevant; recall measures how many relevant items were found.
MRR and nDCG	Ranking metrics that reward placing relevant evidence near the top of the retrieved list.

3 Methodology

This chapter specifies the research design and implementation methodology for an educational chatbot that uses retrieval-augmented generation, a symptom-intervention graph, and nine coordinated software agents to answer questions concerning women’s mental health. The intended contribution is not an automated diagnostic or therapeutic system. It is an evidence-grounded educational system that retrieves traceable information, represents uncertainty, identifies situations that require professional or urgent care, and explains how mental, reproductive, sleep, nutritional, behavioral, and social factors may intersect. This distinction determines the architecture, data selection criteria, agent boundaries, evaluation measures, and safety controls described below.

The methodology follows a comparative experimental approach. A working artifact will be developed iteratively and evaluated against simpler baselines. The central hypothesis is therefore not that multiple agents are inherently superior, but that domain specialization, graph-supported evidence integration, and explicit safety gates can improve retrieval relevance, citation support, coverage of cross-domain symptom patterns, and safe uncertainty communication when compared with a single-agent RAG system. The design will be accepted only if these improvements are demonstrated empirically.

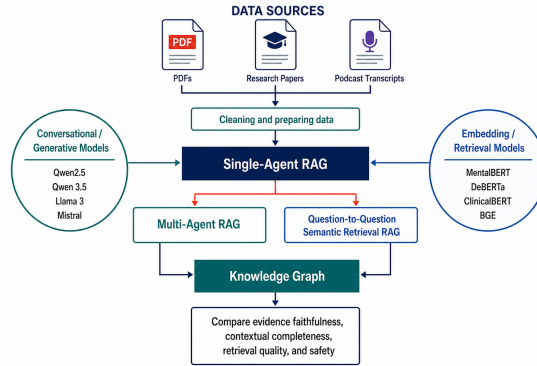


Figure 1: Project design diagram

3.1 Research Questions and Scope

The primary research question investigates how different RAG-centred architectures, including single-agent RAG and multi-agent RAG (with passage retrieval and question-to-question semantic retrieval), compare in producing evidence-faithful, contextually complete, and safety-appropriate educational responses to multi-symptom women’s mental health queries. The secondary research question focuses on adding a knowledge graph and evaluating if it improves cross-domain evidence retrieval, response grounding, and overall answer quality compared with non-graph retrieval approaches. This is illustrated in Figure 1.

The population scope includes diverse range of women, seeking educational information about mood, anxiety, stress, fatigue, sleep, menstruation-related symptoms, pregnancy and postpartum mental health, perimenopause, nutrition, body-related concerns, and lifestyle factors. The system will be inclusive of people for whom reproductive or hormonal information is relevant and will not assume that every woman menstruates or has above mentioned issues. Age, reproductive stage, work pattern, caregiving responsibilities, cultural context, food and exercise access, social support, and self-reported family history may be used as optional context.

The chatbot will use non-diagnostic language such as ‘possible explanations include’, ‘this pattern can be associated with’, and ‘these conditions might be associated with’. It will not prescribe medication, advise medication discontinuation, provide individualized supplement dosing, or present a graph relation as proof of causation. The appropriate research outcomes are thus information-retrieval quality, grounding, safety behavior, calibration, clarity, and educational usefulness rather than clinical diagnostic accuracy.

3.2 Collecting Database

As noted in the introduction, mental health remains one of the most underfunded and under-researched areas of health when compared with many physical health conditions, despite mental disorders currently being reported as the leading cause of global disability. The data collection process for this thesis was limited to resources available through UCL’s licensed access, as well as materials permitted for non-commercial use. Most of the academic literature was obtained from ScienceDirect. These research papers were saved as HTML files and subsequently cleaned and prepared for API-based processing. The selected papers focused on the causes of mental health difficulties and on the relationship between mental health symptoms and broader life domains. These included lifestyle factors, sleep patterns, hormonal health, nutrition, socioeconomic background, meditation, breathing exercises, and related behavioural and environmental factors. In addition to academic publications, this thesis considers podcasts as an increasingly significant source of health-related information. Podcasts have become one of the most popular contemporary formats for disseminating knowledge. As of June 2026, Apple Podcasts alone contained more than 123 million published episodes, while monthly podcast listenership continued to grow, with 42% of adults in the United Kingdom and 58% of adults in the United States reporting that they had listened to a podcast in a month. Given their increasing reach and influence, podcasts are becoming an important subject of academic research. This thesis therefore examines their role in question-to-question semantic retrieval. The podcast transcripts used in this thesis are taken from Huberman Lab, a leading health and fitness podcast on Apple Podcasts and Spotify, and one of the top-ranked podcasts globally. The podcast is hosted by Dr Andrew Huberman, a neuroscientist and tenured professor in the Department of Neurobiology, Psychiatry and Behavioural Sciences at Stanford School of Medicine. The show includes more than 400 episodes, consisting of both solo episodes and interviews with expert guests, many of whom hold doctoral qualifications. Human-reviewed transcripts, prepared by the Huberman Lab team, are available for purchase at a cost of £7.50. These transcripts are used in this thesis strictly for educational and non-commercial purposes.

3.3 Preparing and cleaning data

3.3.1 Science Direct research papers

The ScienceDirect HTML files were not used as the final article text for the RAG corpus. Instead, they were used as an intermediate source of publication metadata. This was necessary because saved HTML webpages contain a large amount of material that is not part of the research article itself, such as menus, scripts, cookies, page layout, advertisements, navigation links, and browser/session-specific data. Therefore, the cleaning process focused on extracting only stable scholarly identifiers and metadata from each saved HTML page. The purpose of this stage was to transform locally saved Science Direct webpages into a structured manifest that could later be used for official Elsevier API retrieval.

The first step was to scan the folder containing the collected resources and identify which files were ScienceDirect HTML pages. A file was treated as a ScienceDirect source if either the filename or the file content contained indicators such as “ScienceDirect” or “sciencedirect.com”. Each relevant HTML file was then read using a tolerant UTF-8 reading method, so that small encoding issues in saved webpages would not interrupt the extraction process. After identifying the ScienceDirect HTML files, the script extracted machine-readable metadata from the page header. Instead of

copying text from the visible webpage, the process looked for structured fields embedded in the HTML, such as DOI, title, canonical article URL, and publisher identifiers. The most important identifiers were the DOI and the PII. The DOI is the permanent digital object identifier of the article, while the PII is Elsevier’s internal publisher item identifier, commonly visible in ScienceDirect article URLs. These identifiers are important because they allow the article to be retrieved through official API endpoints rather than through webpage scraping.

The extracted information was then cleaned and normalised. For example, DOI values were standardised by removing unnecessary URL prefixes, removing trailing punctuation, and encoding special characters so that the DOI could safely be inserted into an API URL. The title was also normalised by removing extra spaces and ScienceDirect page suffixes. If a DOI was not found in one metadata field, the script searched several alternative metadata fields and DOI-like patterns in the HTML. This made the process more robust because saved webpages are not always identical in structure. The output of this cleaning stage was a manifest file. The manifest is a structured catalogue of the ScienceDirect sources. For each article, it records the original saved filename, a short file hash for traceability, article title, DOI, PII, Elsevier EID, canonical ScienceDirect URL, prepared API URL using DOI, prepared API URL using PII, suggested location for saving retrieved XML, API key storage method, retrieval status, and notes. In this project, the manifest contained 47 ScienceDirect HTML records. Of these, 46 were marked as ready for API retrieval because a DOI or PII was found, while 1 record was marked as missing an identifier and requiring manual checking.

The API key itself was not stored inside the manifest. Instead, the key is stored privately as an environment variable called `ELSEVIER_API_KEY`, usually in a local `.env` file. The manifest documents how retrieval should happen, while keeping the private authentication credential separate.

The second stage used the manifest to prepare and perform official Elsevier API calls. For each article, the retrieval script read the DOI and PII endpoints from the manifest. It attempted to retrieve the article through the DOI endpoint first and used the PII endpoint as a fallback. The request included the Elsevier API key in the request header and asked for a structured response format such as XML. If the API call was successful, the returned XML file was saved locally in the project output folder. If the call failed, the failure was recorded in a retrieval log.

The retrieval log makes the process auditable. It records the retrieval date, article title, DOI, PII, endpoint used, request URL, HTTP status code, whether the call succeeded, saved XML path, response size, and error message. This means that the dataset preparation process does not silently ignore failed articles. Instead, every successful and unsuccessful retrieval attempt is documented. For example, a 200 status code means that the article was successfully retrieved, while 400 indicates that the request format or parameter was invalid, 401 usually means an API key or authentication issue, 403 indicates an entitlement or permission issue, and 404 means that the requested identifier was not found.

This workflow was chosen because it separates metadata preparation from full-text retrieval. The saved HTML files are used only to identify the article, while the actual article content is retrieved through the official API when access is permitted. This is more reproducible and more appropriate for academic text and data mining than extracting article text directly from browser-saved pages. Elsevier’s documentation states that ScienceDirect APIs provide access to scholarly content and that API views determine which metadata or full-text fields are returned, depending on access rights and entitlements.

After retrieval, the XML files can be used in the RAG pipeline. The XML content can be parsed into

article sections, such as title, abstract, introduction, methods, results, discussion, and conclusion where available. These sections can then be split into smaller text chunks. Each chunk can be assigned metadata such as source title, DOI, source file, section name, article type, agent domain, symptom tags, concept tags, life-stage tags, and access or license information. These chunks can then be embedded and stored in the vector database. During retrieval, the chatbot can use this metadata to filter or prioritise evidence, for example by selecting chunks related to sleep, hormonal health, nutrition, interventions, safety, or mental health.

3.3.2 Q&A extraction from podcasts

The question-answer dataset was created through a source-grounded, extractive pipeline rather than by asking a language model to invent questions or answers. First, the PDF, that contained all of the copy-pasted transcripts at one place, was converted into page-level text using pypdf, while preserving page numbers so that every resulting pair could be traced to its original location. The transcript was then divided into episode segments using page transitions, episode introductions, guest names and sign-offs. Within each segment, regular expressions identified speaker labels and reconstructed consecutive speaker turns, including turns that continued across page boundaries. A candidate pair was created when an Andrew Huberman turn contained either a question mark or an explicit interview prompt, such as “explain”, “describe” or “tell us”, and the immediately following turn belonged to the expected interviewee. The question was extracted from the final one or two interrogative sentences of the interviewer’s turn, while the answer was taken directly from the beginning of the following speaker turn and had limited number of words. Thus, the answers are transcript-derived excerpts rather than model-generated medical summaries. This design is related to datasets such as SQuAD, in which answers are grounded in identifiable passages, although the present project uses speaker-turn boundaries rather than manually selected answer spans (Rajpurkar et al., 2016). Each pair was supplemented with provenance and routing metadata, including the episode title, question speaker, answer speaker, question type, primary and secondary agents, context dependency, safety flag and review status. Candidate quality was estimated using question and answer length, domain terminology and context independence. Short follow-ups containing unresolved expressions such as “this”, “that” or “it” were penalized. A lexical-diversity procedure then reduced repetitive questions by subtracting the maximum Jaccard similarity to already selected questions from the candidate’s quality score. Context-dependent questions were either removed, minimally rewritten into standalone questions or connected to previous and next sentences when possible, while retaining the original wording in a separate provenance field. This practice is supported by work on conversational question rewriting: CANARD formalizes the conversion of context-dependent questions into self-contained questions that preserve the same answer (Elgohary et al., 2019). The use of answer-alignment checks is also methodologically consistent with round-trip filtering, where a generated question is retained only if an answer model can recover the original answer (Alberti et al., 2019). Following cleaning, the files resulted in 492 retained pairs. The resulting pairs can be embedded for question-to-question retrieval, an approach directly related to PAQ and RePAQ, which demonstrate that a user question can retrieve semantically similar stored questions and return their associated answers (Lewis et al., 2021).

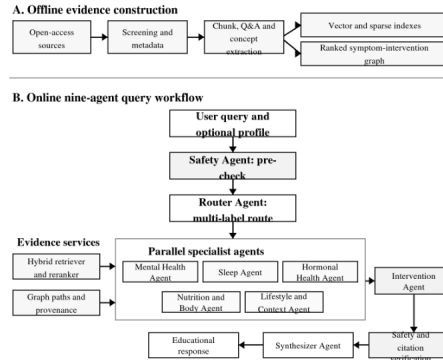


Figure 2: Proposed architecture for the nine-agent graph-supported RAG system. The offline pipeline produces a vector/sparse index and a provenance-rich symptom-intervention graph. During online use, the Safety and Router Agents precede parallel specialist retrieval; the Intervention and Synthesizer Agents operate only on structured, cited outputs, followed by safety and citation verification.

3.4 LLMs that will be used in the thesis

3.5 Single-Agent RAG

The primary research question is focused on exploring different RAG architectures, starting with the baseline, since that will allow better track of possible issues. Single-Agent RAG baseline will combine all of the pdfs from Science Direct and podcast transcript form Huberman Lab and use it with the passage retrieval.

3.6 Proposed Nine-Agent Architecture

The proposed Multi-Agent system contains nine logical agents: Router, Safety, Psychological Symptoms Agent, Sleep, Hormonal Health, Nutrition and Body, Lifestyle and Context, Intervention, and Synthesizer. The Safety Agent is reused at several points but counts as one logical agent. The five domain specialists may run in parallel after routing, while safety, intervention filtering, synthesis, and final verification remain ordered. Figure ?? separates the offline evidence-construction pipeline from the online query workflow.

Table 2: Operational responsibilities of the nine logical agents.

Agent	Bounded responsibility	Structured output
Router	Multi-label domain classification and query rewriting	Agent list, rewritten queries, route confidence
Safety	Crisis, self-harm, eating-risk, abuse, urgent medical, medication and supplement safeguards	Risk level, escalation, allowed answer scope
Psychological Symptoms	Mood, anxiety, trauma/stress, cognition, somatic expression, comorbidity and impairment	Mental-health concepts, evidence, uncertainty

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Table 2 continued

Agent	Bounded responsibility	Structured output
Sleep	Insomnia, hypersomnia, circadian timing, sleep deprivation, fatigue and sleep-related red flags	Sleep pattern and evidence-supported education
Hormonal Health	Cycle timing, PMS/PMDD, pregnancy/postpartum, perimenopause and reproductive context	Timing hypotheses, tracking questions, referral cues
Nutrition and Body	Diet quality, appetite, caffeine, possible deficiencies, body image, eating risk and supplement cautions	Evidence-graded nutrition/body information
Lifestyle and Context	Age, work, caregiving, activity, stress, culture, access, support, substance use and family history	User-provided context modifiers and access-aware options
Intervention	Links supported symptom pathways to low-risk education or clinician-discussion options	Ranked actions, evidence grade, contraindications
Synthesizer	Reconciles agent outputs and writes one concise educational answer	Cited answer, uncertainty, red flags and follow-up

3.7 Evidence-Based Justification for the Agent Domains

3.8 Mental Health as a product of different domains

The proposed multi-agent architecture is grounded in the understanding of mental health as a multifactorial and context-dependent phenomenon rather than the consequence of a single biological, psychological, or social cause.

The choice of agents in this project was informed through a preliminary concept-identification process conducted on a selected subset of the research papers collected for the study. In the first stage, common phrases, symptoms, conditions, lifestyle factors, and contextual determinants were manually extracted from highlighted passages in the source documents. These extracted terms were then organized into a concept matrix, where each phrase was recorded together with its synonyms or closely related expressions. This step was important because mental health concepts are often described using different terminology across sources. For example, depression may appear as depressive symptoms, low mood, or major depressive disorder; anxiety may appear as worry, anxious symptoms, or generalized anxiety disorder; and sleep problems may appear as insomnia, sleep disturbance, sleep deprivation, or poor sleep quality.

The extracted phrases showed that women’s mental health is not limited to psychological symptoms alone, but is connected to several interacting domains. Commonly identified mental health terms included anxiety, depression, stress, burnout, mood changes, fatigue, emotional regulation, trauma, self-esteem, loneliness, social isolation, resilience, suicidal ideation, and quality of life. At the same time, many phrases were related to biological and hormonal processes, including hormones, cortisol,

estrogen, progesterone, menstrual cycle, PMS, PMDD, pregnancy, postpartum depression, prenatal depression, menopause, perimenopause, and hormonal contraception. This supported the inclusion of a Hormonal and Reproductive Health Agent, because the literature repeatedly connects reproductive life stages and hormonal changes with psychological well-being.

The highlighted phrases also included lifestyle and behavioral factors such as sleep, sleep quality, sleep deprivation, diet, nutrition, exercise, physical activity, mindfulness, yoga, cognitive behavioral therapy, social support, and community-based interventions. These terms justified the inclusion of agents focused on sleep, nutrition and body health, and lifestyle or intervention strategies. In addition, several extracted concepts reflected wider social and demographic determinants, including socioeconomic status, financial instability, economic pressures, societal pressures, social media exposure, digital dependency, health care utilization, access to care, gender, sex differences, family context, genetics and social connection. These findings indicate that mental health should be modeled as a broad, biopsychosocial construct rather than as an isolated clinical category.

After the initial extraction stage, manual review was used to examine which concepts commonly appeared together in the collected sources. For example, anxiety and depression frequently appeared alongside sleep disturbance, stress, social isolation, hormonal changes, pregnancy, quality of life, and suicide-related safety concerns. Sleep-related phrases appeared near fatigue, mood, diet, glucose regulation, insulin sensitivity, and mental health. Hormonal and reproductive terms appeared together with depression, anxiety, mood changes, postpartum symptoms, pregnancy, menopause, and quality of life. These observed co-occurrence patterns provided a practical basis for defining the agent structure, because they showed that different domains repeatedly contribute to or contextualize women’s mental health concerns.

A substantial part of the broader source selection was also informed by World Health Organization resources, which describe mental health as being shaped by social, biological, environmental, behavioural, and health-system factors.

The World Health Organization states that mental health is influenced by interacting individual, family, community, social, economic, physical, and structural factors, including poverty, violence, disability, inequality, social support, working conditions, and access to healthcare (WHO, 2014; WHO, 2026). This position is supported by Lund et al. (2018), whose systematic review of 289 reviews found evidence linking mental disorders with demographic, economic, neighbourhood, environmental, social, and cultural determinants. Consequently, the proposed agents should not be interpreted as representing independent causes of mental illness. Instead, they represent analytically distinguishable but interconnected evidence domains through which a user’s symptoms and circumstances can be examined. This decomposition is particularly appropriate for women’s mental health because symptoms such as fatigue, low mood, irritability, impaired concentration, appetite change, anxiety, insomnia, and reduced motivation can occur across psychiatric, sleep-related, reproductive, nutritional, physical, and psychosocial contexts. A multi-agent system therefore provides a mechanism for retrieving evidence from several relevant domains before producing a unified educational response, while avoiding the assumption that the first plausible explanation is necessarily correct.

The Psychological Symptoms Agent is justified as the central domain agent responsible for identifying and contextualising psychological, emotional, cognitive, and behavioural symptom patterns. Depression and anxiety commonly involve multiple dimensions rather than one isolated symptom. For example, the Patient Health Questionnaire includes depressed mood, loss of interest, sleep disturbance, appetite change, fatigue, concentration problems, negative self-evaluation, psychomotor change, and thoughts of self-harm (Kroenke, Spitzer and Williams, 2001).

Similarly, the Generalized Anxiety Disorder scale represents anxiety through excessive worry, difficulty controlling worry, restlessness, irritability, difficulty relaxing, and fear that something negative may happen (Spitzer et al., 2006). Importantly, several of these features overlap with the responsibilities of other proposed agents. Fatigue and concentration difficulties, for example, may accompany depression but may also occur in sleep deprivation, nutritional deficiency, chronic stress, perimenopause, or physical illness.

The Psychological Symptoms Agent should therefore identify clinically meaningful psychological patterns without automatically diagnosing a disorder. Its function is to retrieve evidence about depression, anxiety, stress, trauma, emotional regulation, and related symptoms; identify relevant duration, severity, functional impairment, and co-occurring features; and communicate uncertainty to the other agents. This overlap with other domains is not undesirable duplication. It is precisely the reason that cross-agent retrieval and synthesis are required.

The Sleep Agent is strongly supported by evidence that sleep and mental health are related bidirectionally. A systematic review by Alvaro, Roberts and Harris (2013) found prospective evidence that sleep disturbance may predict later depression or anxiety, while psychological difficulties may also contribute to subsequent sleep disturbance. Sleep should therefore be modelled as both a possible contributor to, and a possible consequence of, mental distress rather than as a simple one-directional cause (Alvaro, Roberts and Harris, 2013). More direct evidence is provided by Scott et al. (2021), whose meta-analysis included 65 randomised controlled trials, 72 interventions, and 8,608 participants. Interventions that successfully improved sleep produced significant improvements in composite mental health outcomes, depression, anxiety, stress, and rumination, with greater sleep improvement associated with greater mental-health improvement (Scott et al., 2021). These findings justify a specialist agent that distinguishes difficulty falling asleep, nighttime awakening, early waking, nightmares, excessive sleepiness, irregular sleep timing, insufficient sleep opportunity, and daytime impairment. Such distinctions are important because a general embedding search may treat all references to “tiredness” as equivalent even though their clinical contexts differ. The Sleep Agent should nevertheless avoid claiming that poor sleep explains all reported symptoms because pain, anxiety, depression, caregiving responsibilities, medication, reproductive symptoms, environmental conditions, and other health problems may themselves disturb sleep.

The Hormonal Health Agent is justified by evidence that reproductive events and cyclical symptom patterns can interact with mental health. This agent should be described more precisely as addressing reproductive and hormonal context, rather than attempting to infer a generic “hormonal imbalance.” Premenstrual syndrome and premenstrual dysphoric disorder are defined by physical, cognitive, behavioural, and affective symptoms that emerge during the late luteal phase and improve following the onset of menstruation. Consequently, temporal recurrence across menstrual cycles is more informative than the mere presence of mood symptoms (Carlini et al., 2022). The agent should therefore consider symptom timing, menstrual regularity, pregnancy and postpartum status, perimenopausal stage, contraceptive use, and whether an existing mental-health condition becomes

worse during a reproductive phase. The importance of reproductive stage is also supported by Badawy et al. (2024), whose systematic review included 17 prospective cohort studies with 16,061 women. Their meta-analysis found that perimenopausal women had a higher risk of depressive symptoms or depression than premenopausal women, with a pooled odds ratio of 1.40, while the comparison between postmenopausal and premenopausal women was not statistically significant (Badawy et al., 2024). This result demonstrates why the chatbot should not make broad statements such as “menopause causes depression.” Risk varies by reproductive stage and may interact with previous depression, vasomotor symptoms, sleep disruption, stressful life events, cultural expectations, and social circumstances. A comprehensive review of women’s reproductive mental health similarly describes reproductive mental health as having interconnected biological, psychological, social, cultural, and political foundations (Howard et al., 2025). The Hormonal Health Agent is therefore needed to recognise reproductive timing and retrieve appropriate evidence, but it must not infer hormone levels, endocrine disorders, or causality from symptoms alone. The Nutrition and Body Agent is justified because diet, appetite, exercise, body image, weight change, and physical symptoms may be associated with psychological well-being, while also appearing as symptoms or consequences of mental-health conditions. Firth et al. (2019) conducted a meta-analysis of 16 randomised controlled trials involving 45,826 participants and found that dietary-improvement interventions produced a modest reduction in depressive symptoms. However, the pooled effect on anxiety was not statistically significant, demonstrating that nutrition evidence should be communicated with appropriate outcome-specific uncertainty (Firth et al., 2019). Physical activity also has a substantial evidence base. A network meta-analysis of 218 randomised studies involving 14,170 participants found that several forms of exercise, including walking or jogging, strength training, yoga, and mixed aerobic exercise, were associated with reductions in depressive symptoms, although effect sizes, certainty, adherence requirements, and suitability varied across interventions and populations (Noetel et al., 2024). These findings justify an agent that retrieves evidence concerning dietary patterns, appetite, body-related concerns, physical activity, fatigue, and possible intervention options. Nevertheless, it should not diagnose nutritional deficiencies from nonspecific symptoms or present food and supplements as replacements for psychological or medical care. Supplement recommendations require particular caution because products may have contraindications, medication interactions, inconsistent formulations, and evidence of very different quality.

The Lifestyle and Context Agent is justified by the extensive evidence that mental health is shaped by the conditions in which people live. Relevant information can include age, employment, financial strain, education, caregiving responsibilities, migration, discrimination, relationship quality, exposure to violence, social isolation, cultural expectations, housing, healthcare access, disability, and available social support. These variables should be handled with care, precision and should be optional for user to share. The WHO identifies social inequality and adverse circumstances as important contributors to unequal mental-health risk (WHO, 2014), while Lund et al. (2018) show that such relationships appear across a large body of systematic-review evidence. Family history can also be included as contextual information, but it must be interpreted probabilistically. Familial vulnerability may reflect combinations of genetic susceptibility, shared environment, learned behaviour, stress exposure, family relationships, and access to resources. It does not mean that a user will inevitably develop the same condition as a relative (van Sprang et al., 2022). The agent must therefore avoid deterministic statements and must not use ethnicity, socioeconomic status, age, or family history as automatic diagnostic labels.

The Intervention Agent is required because identifying a potentially relevant domain is different from selecting an appropriate response. This agent should retrieve and rank evidence about psychoeducation, sleep interventions, physical activity, dietary interventions, stress-management strategies, psychological therapies, social support, and indications for professional assessment. Interventions should be classified by evidence level, target symptom, applicable population, contraindications, accessibility, and whether professional supervision is necessary. For example, evidence supporting exercise for depressive symptoms does not mean that every form of exercise is appropriate for every user, while evidence supporting particular treatments for PMDD cannot be generalised to all menstrual discomfort or mood variation. The agent should consequently prioritise low-risk educational suggestions, explicitly grade uncertainty, and distinguish between self-management possibilities and interventions requiring a qualified healthcare professional.

The remaining agents are primarily justified by architectural and safety requirements. The Router Agent enables multi-label routing because one query may simultaneously concern sleep, mood, menstruation, diet, social stress, and safety. The Safety Agent independently examines the input and proposed response for self-harm, suicide, abuse, severe deterioration, psychosis, dangerous medication or supplement advice, and other situations requiring urgent escalation. The WHO mhGAP Intervention Guide treats self-harm and suicide as priority conditions requiring structured assessment and management, supporting a dedicated safety pathway rather than relying on incidental detection by a general-purpose agent (WHO, 2019).

Finally, Synthesizer Agent combines retrieved evidence, resolves contradictions, removes duplication, preserves source attribution, and produces an answer that clearly separates supported information from uncertain possibilities.

3.8.1 Cross-domain evidence supporting specialization

Sleep disturbance warrants a dedicated agent because its relationship with depression and anxiety is frequently bidirectional. A systematic review by ? found evidence for reciprocal prospective relationships among sleep disturbance, anxiety, and depression, while later reviews continue to treat insomnia as both a symptom and a risk pathway. A Sleep Agent can therefore distinguish sleep-onset difficulty, sleep maintenance, early waking, hypersomnia, circadian disruption, shift work, and daytime impairment, while referring mood and safety implications to the relevant agents. Reproductive timing warrants a Hormonal Health Agent because menstrual, perinatal, and menopausal transitions can change symptom interpretation and risk. PMDD includes affective, cognitive, behavioral, sleep, appetite, and physical symptoms whose cyclical timing and functional impairment are central to recognition (??). A recent comprehensive review concludes that women’s reproductive mental health has complex biological, psychological, social, cultural, and political foundations and requires an interdisciplinary approach (?). The agent must therefore avoid simplistic statements such as ‘hormonal imbalance causes depression’ and instead ask about timing, recurrence, impairment, reproductive stage, medication, and professional assessment.

Nutrition, body experience, and physical activity warrant bounded specialist treatment because evidence strength differs by intervention. A meta-analysis of randomized trials found that dietary improvement produced a small reduction in depressive symptoms, with no statistically significant

pooled effect for anxiety, and emphasized heterogeneity and the need for further study (?). A large network meta-analysis found several exercise forms effective for depression, while the authors also reported variation in certainty and study quality (?). Music therapy and mindfulness have evidence in selected contexts, but effect estimates, intervention definitions, and certainty vary (??). These findings support an Intervention Agent that ranks evidence and distinguishes low-risk education from clinician-led treatment rather than listing every plausible activity as equally established.

3.9 How the Agents differ

The implementation begins with shared schemas and one reusable agent runner, not nine independent chatbots. Each AgentSpec contains a name, system prompt, metadata filters, allowed tools, output schema, minimum evidence requirement, and confidence threshold. The generator is loaded once and reused. The specialist runner performs domain-filtered retrieval, reranking, graph expansion, prompt construction, structured generation, and validation. This architecture makes differences among agents explicit and supports ablation experiments by disabling one specification at a time.

Listing 1: Simplified agent contract and reusable domain runner.

```
class AgentSpec(BaseModel):
    name: str
    domains: list[str]
    system_prompt: str
    tools: list[str]
    min_evidence: int = 2

class AgentOutput(BaseModel):
    concepts: list[str]
    evidence_ids: list[str]
    graph_paths: list[str]
    confidence: float
    uncertainty: list[str]
    safety_flags: list[str]

def run_domain_agent(spec: AgentSpec, state: RAGState) -> AgentOutput:
    candidates = hybrid_retrieve(state.user_query, filters=spec.domains)
    evidence = rerank(state.user_query, candidates)[:TOP_N]
    graph_context = graph_expand(extract_query_concepts(state.user_query))
    raw = generator.with_structured_output(AgentOutput).invoke(
        build_prompt(spec, state, evidence, graph_context)
    )
    return validate_output(raw, allowed_evidence=evidence)
```

Pydantic schemas enforce field types and make missing evidence visible. Each output contains evidence identifiers rather than copied paragraphs, limiting prompt growth and allowing later citation verification. The validator rejects outputs with unknown evidence IDs, out-of-range confidence, prohibited diagnostic wording, or intervention claims outside the agent’s permissions. The same design can be implemented with a custom Python orchestrator or a state-graph framework such as LangGraph.

3.9.1 How are agents connected

LangGraph is a lower-level orchestration framework for constructing stateful workflows from nodes, edges and shared state. In the proposed implementation, each agent or validation function can be represented as a node. Workflow edges determine which component runs next and may contain conditional branches, such as immediate escalation after a high-risk Safety result or parallel activation of several specialist agents. Shared state can contain the query, optional user profile, selected routes, retrieved evidence, graph paths, agent outputs, draft response and safety decision. LangGraph also supports checkpointing, persistence, human review and inspection of state transitions (LangGraph overview; Graph API). The LangGraph workflow graph must be distinguished from the symptom-intervention knowledge graph. The workflow graph controls program execution, for example Router $\rightarrow$ Sleep Agent $\rightarrow$ Synthesizer. The knowledge graph represents domain information, for example insomnia $\rightarrow$ associated-with $\rightarrow$ anxiety. LangGraph may orchestrate calls to the knowledge graph, but it does not automatically create or validate medical relationships.

3.9.2 Shared state and orchestration

All nodes read and update one typed RAGState. Required fields are user_query, optional user_profile, safety_flags, selected_routes, evidence_by_agent, graph_paths, agent_outputs, draft_answer, verified_citations, and final_answer. Safety pre-check and routing are sequential. The selected specialist agents can execute concurrently because each writes to a separate output key. The Intervention Agent waits for all specialists, and the Synthesizer waits for intervention filtering. Retries are permitted only for invalid structured output or temporary model failure; they are not used to pressure an agent into producing an answer when evidence is absent.

Listing 2: Online orchestration in implementation-oriented pseudocode.

```
def answer_query(query: str, profile: UserProfile | None) -> FinalAnswer:
    state = RAGState(user_query=query, user_profile=profile)
    state.safety_flags = safety_agent.precheck(state)
    if state.safety_flags.requires_immediate_escalation:
        return safety_agent.crisis_response(state)

    state.selected_routes = router_agent.route(state)
    state.agent_outputs = run_selected_specialists_in_parallel(state)
    state.interventions = intervention_agent.rank_and_filter(state)
    state.draft_answer = synthesizer_agent.compose(state)
    state.verified_citations = verify_claim_support(state.draft_answer)
    return safety_agent.postcheck_and_release(state)
```

3.10 References

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4 Results and Evaluation

- Analysis of results
- Comparison of different approaches
- Limitations and extensions of current work

5 Conclusion